

Important Terms Used in the Chapter

Machine: A device that enables work to be performed in an easier way.

Input Work: The work done on the machine (input energy).

Output Work: The work done by the machine (output energy).

Efficiency: The ratio of the work done by the machine to the work done on the machine, expressed as a percentage.

Effort (E): The force applied to a simple machine to perform work.

Load (L): The force applied by a simple machine on an object to carry out the required work.

Fulcrum: The fixed point about which a rigid beam or body can freely rotate.

Effort Arm: The distance between the fulcrum and the point where effort is applied.

Load Arm: The distance between the fulcrum and the point where the load acts.

Mechanical Advantage (MA): The ratio of the load to the effort applied.

Class I Lever: A lever in which the fulcrum lies between the effort and the load.

Class II Lever: A lever in which the load lies between the fulcrum and the effort.

Class III Lever: A lever in which the effort lies between the fulcrum and the load.

Inclined Plane: A flat surface tilted at an acute angle to the horizontal plane.

Screw: A cylinder or nail surrounded by an inclined plane wound as a spiral groove (thread).

Thread: The inclined plane wrapped spirally around the nail or cylinder of a screw.

Wedge: A tool made of two inclined planes meeting to form a sharp edge.

Pulley: A circular disc with a grooved rim mounted on an axle through which a rope or cable passes.

Block & Axle: The axle is the central spindle around which the disc rotates; the block is the support frame holding the axle.

Block and Tackle: A combination system of several pulleys used to lift heavy loads.

Wheel and Axle: A machine consisting of two concentric cylinders of different diameters that rotate together about a common axis.

Important Mathematical Formulas

Efficiency (η): $\eta = (\text{Output Work} / \text{Input Work}) \times 100$

Efficiency loss: If efficiency is $x\%$, then $(100 - x)\%$ is lost in overcoming friction and other losses.

Mechanical Advantage (MA): $MA = \text{Load} / \text{Effort} = L / E$

Law / Principle of Lever: $\text{Effort} \times \text{Effort Arm} = \text{Load} \times \text{Load Arm}$

Mechanical Advantage of an Inclined Plane: $MA = l / h$

Combined Inclined Plane Equilibrium Relation: $E \times l = L \times h$

Single Fixed Pulley: $MA = 1$

Single Movable Pulley: $MA = L / E = 2 / 1 = 2$

Mechanical Advantage of Wheel and Axle: $MA = \text{Radius of effort wheel} / \text{Radius of load axle}$, when effort is applied to the wheel.

Note: In the inclined-plane formula, l is the sloping length and h is the vertical height. The wheel-and-axle formula depends on where the effort and load are applied.

Daily Life Examples

Machine Type	Common Examples	Functional Role / Purpose
Simple Tools	Potato peeler, kitchen knife	Cuts down required effort to prepare food.
Class I Lever	See-saw, pair of scissors	Fulcrum sits in the middle; scissors cut material by balancing opposed forces.
Class II Lever	Wheelbarrow, nutcracker	Load is centered between fulcrum and effort; offers $MA > 1$ for heavy lifting/cracking.
Class III Lever	Fishing rod, forceps/tweezers	Effort is centered; provides speed and precision ($MA < 1$).
Inclined Plane	Parking ramp, playground slide, dump truck bed	Reduces effort when elevating heavy barrels or vehicles along a slope.
Screw	Screws, automobile jack	Converts rotational force to linear lift or creates a firm fastening grip.
Wedge	Axe, saw, knife blade	Splits timber, cuts wood, and slices objects cleanly.
Fixed Pulley	Well bucket rope, flagpole mechanism	Redirects downward pulling effort into upward motion.
Movable Pulley / System	Construction crane, block and tackle rigs	Halves effort or compounds mechanical advantage to hoist heavy construction loads.
Wheel and Axle	Door knob, screwdriver, water tap	Small force applied on outer perimeter easily turns the inner axle to rotate screws or open valves.